FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 08

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Recap *of previous lecture*

(AP/GP/AGP)



Topics to be covered

$${}^nC_r = {}^nC_{n-r}$$

" BINOMIAL THEOREM "

$$\boxed{{}^nC_r = \frac{n!}{r!(n-r)!}}$$

eg ${}^{11}C_3 = \frac{11!}{3!\,8!} = \frac{11\times10\times9}{3\times2\times1}$

eg ${}^8C_2 = \frac{8!}{2!\,6!} = \frac{8\times7}{2\times1}$, eg ${}^5C_1 = \frac{5}{1} = 5$

eg ${}^6C_0 = \frac{6!}{0!\,6!} = 1$, ${}^{11}C_8 = ? = {}^{11}C_3 = \frac{11\times10\times9}{3\times2\times1}$



(*) P.T; $\boxed{{}^nC_r = {}^nC_{n-r}}$

$$LHS = \frac{n!}{r!\,(n-r)!} \quad \text{---(1)}$$

$$RHS = \frac{n!}{(n-r)!\,\left(n-(n-r)\right)!}$$

$$= \frac{n!}{(n-r)!\,r!} \quad \text{---(2)}$$

By (1) & (2); LHS = RHS

i.e $\boxed{{}^nC_r = {}^nC_{(n-r)}}$

(2) $\boxed{{}^nC_0 = 1,\ {}^nC_1 = n}$, $\boxed{{}^nC_n = 1}$

eg ${}^6C_0 = 1,\ {}^6C_1 = 6,\ {}^6C_6 = 1$

${}^3C_0 = 1,\ {}^3C_1 = 3,\ {}^3C_2 = {}^3C_{3-2} = {}^3C_1 = 3$

eg ${}^4C_2 = \frac{4\times3}{2\times1} = 6,\ {}^4C_3 = ? = {}^4C_1 = 4$

$(x+a)^1 = x+a$

$(x+a)^2 = x^2+2ax+a^2$

$(x+a)^3 = x^3+3x^2a+3xa^2+a^3$

$(x+a)^4 = (x+a)^2(x+a)^2$

$= (x^2+2ax+a^2)(x^2+2ax+a^2)$

$= 1x^4+4x^3a+6x^2a^2+4xa^3+1a^4$

$= {}^4C_0x^4+{}^4C_1x^3a+{}^4C_2x^2a^2+{}^4C_3xa^3+{}^4C_4a^4$

$(x+a)^1 = 1x+1a = {}^1C_0x+{}^1C_1a$

$(x+a)^2 = 1.x^2+2xa+1.a^2 = {}^2C_0x^2+{}^2C_1xa+{}^2C_2a^2$

$(x+a)^3 = 1.x^3+3x^2a+3xa^2+1.a^3$

$= {}^3C_0x^3+{}^3C_1x^2a+{}^3C_2xa^2+{}^3C_3a^3$

Ie in General, $(x+a)^n = {}^nC_0x^n+{}^nC_1x^{n-1}a+{}^nC_2x^{n-2}a^2+{}^nC_3x^{n-3}a^3+\cdots+{}^nC_rx^{n-r}a^r+\cdots+{}^nC_nx^0a^n$

$$(x+a)^n = \sum_{r=0}^{n} {}^nC_r x^{n-r}a^r$$

& $(r+1)^{th}$ term $= T_{r+1} = {}^nC_rx^{n-r}a^r$

BINOMIAL Exp for any +ve integral power $\rightarrow$ i.e n = +ve integer

$$(x+a)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1}a + {}^nC_2 x^{n-2}a^2 + \cdots + {}^nC_r x^{n-r}a^r + \cdots + {}^nC_n a^n$$

$$(x+a)^n = \sum_{r=0}^{n} {}^nC_r x^{n-r} a^r \quad \& \quad (r+1)^{th} \text{ term} = T_{r+1} = {}^nC_r x^{n-r} a^r$$

(1) Sum of the indices in each term $= n$

(2) Total Number of terms in the B. Exp of $(x+a)^n = n+1$

(3) Here ${}^nC_0, {}^nC_1, {}^nC_2, {}^nC_3, \ldots, {}^nC_n$ are called Binomial coeff.

(4) Middle term in the B. Exp^{n}

Case I: if n = even then Middle term $= \left(\frac{n}{2}+1\right)^{th}$ term. eg $(x+a)^4$ the M. term $= 3^{rd}$ term

Case II: if n = odd then there will be two Middle terms and these are $\left(\frac{n+1}{2}\right)^{th}$ term & $\left(\frac{n+3}{2}\right)^{th}$ term.

for eg in $(x+a)^5$; $n=5$ so Middle term $= 3^{rd}$ & 4th term.

PASCAL TRIANGLE – Binomial Coeff follow the Pascal Triangle as follows;

BINOMIAL Term	Coeffs of Various terms.
$(x+a)^0$	1
$(x+a)^1$	1 – 1
$(x+a)^2$	1 – 2 – 1
$(x+a)^3$	1 – 3 – 3 – 1
$(x+a)^4$	1 – 4 – 6 – 4 – 1
$(x+a)^5$	1 – 5 – 10 – 10 – 5 – 1
$(x+a)^6$	1 – 6 – 15 – 20 – 15 – 6 – 1

$$(x+a)^6 = 1x^6 + 6x^5a + 15x^4a^2 + 20x^3a^3 + 15x^2a^4 + 6xa^5 + 1a^6$$

Q: 13th term in the expansion of $\left(9x-\frac{1}{3\sqrt{x}}\right)^{18}$ = ?

Sol: $T_{r+1} = {}^{n}C_r \, x^{n-r} \cdot A^r$

$$T_{13} = T_{12+1} = {}^{18}C_{12}(9x)^{18-12}\left(\frac{-1}{3\sqrt{x}}\right)^{12}$$

$$= {}^{18}C_{12}\, 9^6 \cdot x^6 \times \frac{1}{3^{12}\left(x^{\frac{12}{2}}\right)}$$

$$= {}^{18}C_{12}\frac{(3^2)^6 \cdot x^6}{3^{12}\, x^6} = {}^{18}C_{12}$$

Q: Find 11th term from end in $\left(2x-\frac{1}{x^2}\right)^{25}$

Total terms $(n) = 26$

then n^{th} term from end $= \left[m-(n-1)\right]^{th}$ term from beginning

1, 2, 3, ... 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26

11th to 9 8 7 6 5 4 3 2 1

Now $m=26$, $n=11 \Rightarrow (m-n+1) = 16$

11th term from end = 16th term from beginning

$$= T_{16} = T_{15+1} = {}^{25}C_{15}(2x)^{25-15}\left(\frac{-1}{x^2}\right)^{15}$$

$$= -{}^{25}C_{15}\, 2^{10}/x^{20}$$

Q: Find the term independent from x in the Expansion of $\left(x-\frac{1}{x}\right)^{12}$

Sol: Let $(r+1)^{th}$ term is Ind of x.

$$T_{r+1} = {}^{n}C_r \, X^{n-r} \, A^r$$

$$= {}^{12}C_r (x)^{12-r} \left(-\frac{1}{x}\right)^r$$

$$= {}^{12}C_r (-1)^r \, x^{12-2r} \quad \text{---(1)}$$

ATQ, $12-2r=0 \Rightarrow r=6$

i.e 7th term will be Ind of x & $T_7 = {}^{12}C_6$ Ans

Q: Find the coeff of x^{40} in the Expansion of $(1+2x+x^2)^{27}$?

Sol: $(1+2x+x^2)^{27} = \left[(1+x)^2\right]^{27} = (1+x)^{54}$

$\because (1+x)^{54} = \sum_{r=0}^{54} {}^{54}C_r \, x^r$

$\therefore$ Coeff of $x^{40} = {}^{54}C_{40} = {}^{54}C_{14}$ Ans

Special Case – W.k.t that $(x+a)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1}a + {}^nC_2 x^{n-2}a^2 + \cdots + {}^nC_{n-1} x a^{n-1} + {}^nC_n a^n$

Put $a=1$; then $\boxed{(x+1)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} + {}^nC_2 x^{n-2} + \cdots + {}^nC_{n-1} x + {}^nC_n \cdot 1}$

$\because {}^nC_r = {}^nC_{n-r}$ then $(x+1)^n = {}^nC_n x^n + {}^nC_{n-1} x^{n-1} + {}^nC_{n-2} x^{n-2} + \cdots + {}^nC_1 x + {}^nC_0 \cdot 1$

n = +ve int. or $\boxed{(1+x)^n = {}^nC_0 \cdot 1 + {}^nC_1 \cdot x + {}^nC_2 x^2 + \cdots + {}^nC_{n-1} x^{n-1} + {}^nC_n x^n}$

Let it is in the form of Increasing Powers of x.

Note – ① $\boxed{T_{r+1} = {}^nC_r x^r}$ & Coeff of $(r+1)^{th}$ term $= {}^nC_r$

② In the Binomial expansion of $(1+x)^n$ the coeffs that are equidistant from beginning and end are always equal. $\because ({}^nC_r = {}^nC_{n-r})$ & coeff of $x^r = {}^nC_r$

Eg: Which one is Larger ? $(1.01)^{1000000}$ & 10000

Sol: $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \cdots + {}^nC_n x^n$

$(1.01)^{1000000} = (1+0.01)^{1000000}$, Here $n = 1000000$ & $x = 0.01$

$= {}^{1000000}C_0 + {}^{1000000}C_1 (0.01) + \underbrace{{}^{1000000}C_2 (0.01)^2 + \cdots + {}^{1000000}C_{1000000} (0.01)^{1000000}}$

$= 1 + 1000000\left(\frac{1}{100}\right) + \text{+ve Quantity}$

$(1.01)^{1000000} = 10000 + (1 + \text{+ve Quantity}) = 10000 + (\text{+ve No})$

So $(1.01)^{1000000} - 10000 = \text{+ve No.}$ > 0

So $(1.01)^{1000000}$ is greater

Properties of Binomial Coeff. :-

(1) Sum of B. Coeffs $= 2^n \Rightarrow$ $\boxed{{}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n}$

Proof: W.K that ${}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \dots + {}^nC_n x^n = (1+x)^n$

Put $x = 1$, ${}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n = 2^n$ //

(2) Sum of odd B. Coeffs = Sum of Even B. Coeffs $= 2^{n-1}$

$\Rightarrow$ $\boxed{{}^nC_1 + {}^nC_3 + {}^nC_5 + \dots = {}^nC_0 + {}^nC_2 + {}^nC_4 + \dots = 2^{n-1}}$

(3) $\boxed{{}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + {}^nC_4 - {}^nC_5 + \dots + (-1)^n {}^nC_n = 0}$

(4) Sum of the Sq. of B. Coeffs $= {}^{2n}C_n \Rightarrow$ $\boxed{({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2 = {}^{2n}C_n}$

Proof of 2nd → w.k.that $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + {}^nC_4 x^4 + \dots + {}^nC_n x^n$

Put $x=1$; $\quad 2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + {}^nC_4 + \dots + {}^nC_n$

again Put $x=-1$; $\quad 0 = {}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + {}^nC_4 + \dots + (-1)^n \cdot {}^nC_n$

Adding, $\quad 2^n = 2\,{}^nC_0 + 2\,{}^nC_2 + 2\,{}^nC_4 + \dots$

$\therefore \quad {}^nC_0 + {}^nC_2 + {}^nC_4 + \dots = \frac{2^n}{2} = 2^{n-1}$ //

Now Subtracting; $\quad 2^n = 2\,{}^nC_1 + 2\,{}^nC_3 + 2\,{}^nC_5 + \dots$

i.e. $\quad {}^nC_1 + {}^nC_3 + {}^nC_5 + \dots = \frac{2^n}{2} = 2^{n-1}$

Proof of 4th part:- we know that, ${}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \dots + {}^nC_n x^n = (1+x)^n$ ——— (1)

also we can write, ${}^nC_0 x^n + {}^nC_1 x^{n-1} + {}^nC_2 x^{n-2} + \dots + {}^nC_{n-1} x + {}^nC_n 1 = (x+1)^n$ ——— (2)

Multiplying; $\left[{}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n\right]\left[{}^nC_0 x^n + {}^nC_1 x^{n-1} + {}^nC_2 x^{n-2} + \dots + {}^nC_n 1\right] = (1+x)^{2n}$

Comparing the coeff. of x^n both sides;

$$({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2 = {}^NC_r$$

$N = 2n$

$r = n$

i.e. $({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2 = {}^{2n}C_n$

Q: the sum of the last 10 coeff. in the expansion of $(1+x)^{19}$ when expanded in the ascending powers of x

Sol: $(1+x)^{19} = {}^{19}C_0 + {}^{19}C_1 x + {}^{19}C_2 x^2 + {}^{19}C_3 x^3 + \ldots + {}^{19}C_9 x^9 + {}^{19}C_{10} x^{10} + {}^{19}C_{11} x^{11} + \ldots + {}^{19}C_{19} x^{19}$

10 terms. Last 10 terms.

Let, $S = {}^{19}C_{10} + {}^{19}C_{11} + {}^{19}C_{12} + \ldots + {}^{19}C_{18} + {}^{19}C_{19}$

$S = {}^{19}C_9 + {}^{19}C_8 + {}^{19}C_7 + \ldots + {}^{19}C_1 + {}^{19}C_0$ $\left(\because {}^{n}C_r = {}^{n}C_{n-r}\right)$

$+$

$2S = {}^{19}C_0 + {}^{19}C_1 + {}^{19}C_2 + \ldots + {}^{19}C_9 + {}^{19}C_{10} + {}^{19}C_{11} + \ldots + {}^{19}C_{19}$

$2S = 2^{19} \Rightarrow S = 2^{18}$ Ans

Q: Evaluate $^{20}C_0 + {}^{20}C_1 + {}^{20}C_2 + \dots + {}^{20}C_9 + {}^{20}C_{10} = ?$

Sol: Let $S = {}^{20}C_0 + {}^{20}C_1 + {}^{20}C_2 + \dots + {}^{20}C_9 + {}^{20}C_{10}$

again $S = {}^{20}C_{20} + {}^{20}C_{19} + {}^{20}C_{18} + \dots + {}^{20}C_{11} + {}^{20}C_{10}$ $\left(\because {}^nC_r = {}^nC_{n-r}\right)$

$+$

$$2S = {}^{20}C_0 + {}^{20}C_1 + {}^{20}C_2 + \dots + {}^{20}C_{10} + {}^{20}C_{11} + {}^{20}C_{12} + \dots + {}^{20}C_{19} + {}^{20}C_{20} + {}^{20}C_{10}$$

$$2S = 2^{20} + {}^{20}C_{10}$$

$$S = 2^{19} + \frac{{}^{20}C_{10}}{2}$$ Ans

BINOMIAL Th for +ve integral index → $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n$

Here Total term = $n+1$ & n = +ve integer

BINOMIAL Th for any index → Let n = −ve int or Fraction then B.Exp of $(1+x)^n$ contains Infinite No. of terms & it is given as

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots \infty \text{ terms} \quad |x| < 1$$

Eg $(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \cdots \infty$ terms

$(1-x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \cdots \infty$ terms $= \frac{1}{1-x}$ = Sum of ∞ G.P

$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \cdots \infty$ terms. $= \sum_{r=0}^{\infty} (r+1)x^r$

$(1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + 15x^4 + \cdots \infty$ terms. $= \sum_{r=0}^{\infty} \left(\sum r+1\right) \cdot x^r$

$\Sigma(n) = \frac{n(n+1)}{2}$

$\Sigma 2 = \frac{2\times3}{2} = 3 = 1+2$

$\Sigma 3 = \frac{3\times4}{2} = 6 = 1+2+3$

$\Sigma 4 = \frac{4\times5}{2} = 10 = 1+2+3+4$

$\Sigma 5 = \frac{5\times6}{2} = 15 = 1+2+3+4+5$

Note In the expression $(x+a)^n$, if n = -ve int or fraction then process of expanding it, is as follows;

(i) if $x < a \Rightarrow \frac{x}{a} < 1$

$(x+a)^n = a^n\left[1+\frac{x}{a}\right]^n = a^n\left[\text{use } (1+x)^n\right]$

(ii) if $x > a$ then $\frac{a}{x} < 1$

$(x+a)^n = x^n\left[1+\frac{a}{x}\right]^n = x^n\left[\text{use } (1+x)^n\right]$

Q the coeff of x^n in the expansion of $(1-x)^{-2}$ will be?

Ans: Coeff of $x^n = (n+1)$

Q. the coeff of x^n in the Exp $(1-2x+3x^2-4x^3+5x^4-\ldots\ldots)^{-n}$

$$= \left[(1+x)^{-2}\right]^{-n} = (1+x)^{2n}$$

So Coeff of x^n in the Exp of $(1+x)^{2n} = {}^{2n}C_n$ Ans

Note:-

(i) $\boxed{{}^nC_r = {}^nC_{n-r}}$

(ii) $\boxed{\dfrac{{}^nC_{r+1}}{{}^nC_r} = \dfrac{n-r}{r+1}}$

(iii) $\boxed{\dfrac{{}^nC_r}{{}^nC_{r-1}} = \dfrac{n-r+1}{r}}$

(ii) $\dfrac{{}^nC_{r+1}}{{}^nC_r} = \dfrac{\frac{n!}{(r+1)!\,(n-r-1)!}}{\frac{n!}{r!\,(n-r)!}} = \dfrac{n!}{(r+1)!(n-r-1)!} \times \dfrac{r!(n-r)!}{n!} = \dfrac{r!\,(n-r)\cdot(n-r-1)!}{(r+1)r!\,(n-r-1)!} = \dfrac{n-r}{r+1}$

Q: In a B.Exp of $(1+x)^n$, the coeffs of 5th, 6th & 7th term are in AP then Find $n=?$

Sol: $(1+x)^n = \sum_{r=0}^{n} {}^nC_r x^r$ i.e $\boxed{T_{r+1} = {}^nC_r x^r}$ So Coeff of $(r+1)^{th}$ term $= {}^nC_r$

Now $T_5 = T_{4+1} = {}^nC_4 x^4$

$T_6 = T_{5+1} = {}^nC_5 x^5$

$T_7 = T_{6+1} = {}^nC_6 x^6$

$\Rightarrow {}^nC_4, {}^nC_5, {}^nC_6$ are in AP So.

$2({}^nC_5) = {}^nC_4 + {}^nC_6$

$2 = \frac{{}^nC_4}{{}^nC_5} + \frac{{}^nC_6}{{}^nC_5}$

$\left(\because \frac{{}^nC_{r+1}}{{}^nC_r} = \frac{n-r}{r+1} \right)$

$$2 = \frac{1}{\left(\frac{{}^{n}C_5}{{}^{n}C_4}\right)} + \frac{{}^{n}C_6}{{}^{n}C_5}$$

$$\frac{{}^{n}C_{r+1}}{{}^{n}C_r} = \frac{n-r}{r+1}$$

$$2 = \frac{1}{\frac{n-4}{5}} + \frac{n-5}{6}$$

$$2 = \frac{5}{n-4} + \frac{n-5}{6}$$

$\Rightarrow$ ..... $n = 7, 14$


Thank
YOU